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Vector Spaces

Basis
- Set of LI vectors that span the corresponding space.

e.g. can a VS have more than 1 basis? yes.

The basis is the minimal set.

$V = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix}$ vector space represented by $k \cdot \vec{v}$ where k is a scalar, Dimension of that VS: 1

Dimension
- # of vectors in the basis of the corresponding vector space.

E.g. For $\mathbb{R}^2$, dim = 2

For VS spanned by $\left\{\begin{pmatrix}1\\0\end{pmatrix}, \begin{pmatrix}0\\1\end{pmatrix}, \begin{pmatrix}1\\0\end{pmatrix}\right\}$ → Dim = 2, Basis = $\left\{\begin{pmatrix}1\\0\end{pmatrix}, \begin{pmatrix}0\\1\end{pmatrix}\right\}$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 2 & 3 & 1 & 7 \\ 3 & 4 & 2 & 10 \end{array}\right]$$

→ RREF: $\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right]$

Dim of colm space = 2
(as there are 2 pivot columns, 2 columns are LI in the matrix, so whatever colm space they span, dim = 2)

soln to system $= \begin{pmatrix} 2-2t \\ 1+t \\ t \end{pmatrix} \rightarrow \underbrace{\begin{pmatrix}2\\1\\0\end{pmatrix} + t\begin{pmatrix}-2\\1\\1\end{pmatrix}}_{\text{can this be vector space?}} \in \mathbb{R}^3$

→ no, there's no zero vector.

Soln space for a non-homogeneous system of lin. equations can NOT be a vector space.

$Ax = 0$ $\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 2 & 3 & 1 & 0 \\ 3 & 4 & 2 & 0 \end{array}\right]$

soln $= t\begin{bmatrix}-2\\1\\1\end{bmatrix}$

→ colm space is vector space, dim = 1

Soln space for a homogeneous system of lin. equations is a vector space.

Question 1.

(MCQ) Let A be any $n \times m$ matrix, where $m > n$. Which of the following statements is/are TRUE about the system of linear equations $Ax = 0$?

a) There exist at least $m - n$ linearly independent solutions to this system

b) There exist $m - n$ linearly independent vectors such that every solution is a linear combination of these vectors

c) There exists a non-zero solution in which at least $m - n$ variables are 0

d) There exists a solution in which at least n variables are non-zero

$n \times m \quad m > n$

$\begin{bmatrix} c_1 & c_2 & \cdots & c_m \end{bmatrix}$

a) soln space:

of free vars = # of LI vectors in soln space.

If n LI in col^m space → $m-n$ free vars

If # of LI vectors $< n$ → # of free vars $> m-n$ (as dim of basis of col^m space $< n$) ✓

b) about soln space as well.

exactly $m-n$ LI vectors. NO. can be atleast $m-n$, but can be more.

c) about soln space:

atleast $m-n$ vars are zero.

$\begin{pmatrix} 1 & 0 & 0 & 2 & 4 \\ 0 & 1 & 0 & 3 & 5 \\ 0 & 0 & 1 & 4 & 6 \end{pmatrix}_{3\times5} \begin{pmatrix} a \\ b \\ c \\ d \\ e \end{pmatrix}_{n\times m} = 0$

3 pivot cols, free cols

For $Ax = 0$, we can have 4 non-zero vars. False. only 1 zero var.

take 1 free as non-zero, others zero

3 pivot 1 free

$(n+1)$ non-zero soln^{n}s at max ✗ if n is # of LI vars.

Atleast $m-n-1$ zero vars is true.

$\begin{pmatrix} 1 & 0 & 0 & 2 & 4 \\ 0 & 1 & 0 & 3 & 5 \\ 0 & 0 & 1 & 4 & 6 \end{pmatrix}$ a b c d e

$Ax = 0$

one soln: $d = 0$, $a = 2$, $b = 3$, $c = 4$, $e = 1$ ✓

If we make both $d = e = 0$, we have to make $a = b = c = d = e$ but we are discussing non-zero soln^{n}s.

d) Atleast n non-zero vars:

No.

we could have # of LI col^{m}s $< n$.

$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 3 & 4 & 5 \end{pmatrix}_{3\times5}$ p f

of non-zero vars = 1 pivot + 1 free $= 2 < n = 3$ ✗

Q. Is $\mathbb{R}^m$ is a subspace of $\mathbb{R}^n$? Given that $m < n$?

→ $\mathbb{R}^2$ is subspace of $\mathbb{R}^3$? No.

NO. Dim of each vector in the subspace has to be same as dim of each vector in the original space.

Q. Dim of zero vector space?

→ Zero

$\left\{ k \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \right\}$ is $\left\{ \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \right\}$. Basis is $\{\}$.

↑ Trivial vector space.

Column space
- Vector space spanned by the columns of a matrix is its column space.

Dim = Rank

Null space
- Soln space for Ax=0 is the null space of A.

Dim = Nullity

Rank
- The rank of a matrix A is the dimension of column space of A.

Rank (A) = dim(colm space)
= # of LI colms
= # of pivots in RREF.

Nullity
- The nullity of A is the dim of null space of A.

Nullity (A) = Dim(null space of A)
= # of free vars

Rank-Nullity theorem
Let A be an mxn matrix. Then,

rank(A) + nullity(A) = n

$$A = \begin{matrix} a & b & c \end{matrix} \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 1 & 3 & 1 \end{pmatrix}$$

$$\rightarrow \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix} \rightarrow \begin{matrix} \bar{a} & \bar{b} & \bar{c} \end{matrix} \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{matrix}$$

Rank = 2, Nullity = 1

p f

Null space

soln to $Ax = 0 = \begin{pmatrix} 2t \\ -t \\ t \end{pmatrix} = t\begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$

of free vars = 1

What is colm space of A?

a) span(a, b, c) ✓ b) span($\bar{a}$, $\bar{b}$, $\bar{c}$) c) span(a, b) ✓ d) span($\bar{a}$, $\bar{b}$)

- In row-echelon form, if there are some free vars, then the corresponding colms in the original matrix are dependent colms.
- In row-echelon form of A, if there are some pivot variables, the corresponding colms in A are independent colms.

↗ They can constitute basis for colm space.

so span(a, b) is also the colm space as z is the free var.

NOTE: colm space ≠ span($\bar{a}$, $\bar{b}$)
space can change after reduction.

Full rank matrix (mxn) rank(A) = min(m, n)

Square matrix: If rank(A) = # of colms.

Rectangular matrix: If rank(A) = min(m, n)

Q. what is dim of null space for $Ax=0$ if A is full-rank ?

→ 0 (all col^m vectors of A will be LI, only trivial sol^n is possible).

Q. In $\mathbb{R}^n$, if all vectors lie in $\mathbb{R}^{n-1}$ dimensional subspace, they are LI. Prove this.

Reduced row echelon form

- it has to be in row echelon form.
- all leading entries should be 1.
- all entries above and below the leading entries should be 0.

e.g. $\begin{bmatrix} 1 & 0 & 7 \\ 0 & 1 & 4 \end{bmatrix}$

Rank of augmented matrix

rank = # LI column vectors

$x+y=2$
$x+2y=3$ $\longrightarrow$ $\left(\begin{array}{cc|c} 1 & 1 & 2 \\ 1 & 2 & 3 \end{array}\right)_{2\times 3}$ $\rightarrow$ $\left(\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & 1 & 1 \end{array}\right)$ echelon form

2 pivots. rank = 2 here rank(A) = rank(A|b) = # of variables = n

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$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & -1 & 2 \end{array}\right)$ rank(A) = rank(A|b) = n = 3 unique soln.

$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & 0 & 5 \end{array}\right)$ $R(A)=2$, $R(A|b)=3$ Rank(A|b) ≠ Rank(A)

→ Inconsistent ⇒ No soln.

A: $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$ P P F, r = 2

A|b: $\left(\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & 0 & 5 \end{array}\right)$ P P F P, r = 3

$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{array}\right)$ P P F

for A

P P F F

for (A|b)

$r(A)=2$, $r(A|b)=$ $r(A) = r(A|b) \neq n$ Free var and consistent. ∞ly many soln's.

For $Ax=0$, rank(A) = rank(A|b) = rank(A|0)

Row space

- The space spanned by row vectors of a matrix is its row space.
- Row rank = # of LI row vectors

$$\begin{matrix} R_1 \\ R_2 \\ R_3 \end{matrix}\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix}\begin{matrix} r_1 \\ r_2 \\ r_3 \end{matrix}$$

$C_1\ C_2\ C_3$ $\qquad$ $rC_1\ rC_2\ rC_3$

colm space = $\text{span}(C_1, C_2, C_3) = \text{span}(C_1, C_2) \neq \text{span}(rC_1, rC_2)$

Basis for colm space = (C_1, C_2)

Row space = $\text{span}(R_1, R_2, R_3) = \text{span}(R_1, R_2) = \text{span}(r_1, r_2)$

basis for row space = (R_1, R_2) OR (r_1, r_2)

So row space isn't changed by row reduc^n operat^{n}s.

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

basis of colm space $= \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ $\qquad$ $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$

basis for row space $= (1\ 2\ 3)$ $\qquad$ $(1\ 2\ 3)$

- Row space of A = Colm space of transpose(A)

Rank

Rank = # of LI colms = # of LI rows.

- Rank of matrix A is the # of LI colms (or rows)
- Rank of matrix A is the dimension of colm space (or row space).

Reason: # of pivots doesn't change ib you see either from row/colm perspective.

Row space ≠ colm space

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 1 & 1 \\ 1 & 3 & 4 & 5 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

colm space $\in \mathbb{R}^3$
row space $\in \mathbb{R}^4$

$A_{m \times n} \longrightarrow$ colm space $\in \mathbb{R}^m$

row space $\in \mathbb{R}^n$

Left null space

- Left null space (A) = Null space (transpose A)
- It is the set of all vectors y s.t. $A^Ty=0$

Null space	Left null space
- soln to $Ax=0$	soln to $A^Ty=0$

$$\begin{matrix} R_1 \\ R_2 \\ R_3 \end{matrix}\begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 1 & 1 \\ 1 & 3 & 4 & 5 \end{matrix}\begin{matrix} \\ \\ \\ \end{matrix}\begin{matrix} \\ \\ \\ \end{matrix}\begin{matrix} c_1 & c_2 & c_3 & c_4 \end{matrix}$$

$$\longrightarrow \begin{matrix} R_1' \\ R_2' \\ R_3' \end{matrix}\begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{matrix}\begin{matrix} \\ \\ \\ \end{matrix}\begin{matrix} c_1' & c_2' & c_3' & c_4' \end{matrix}$$

$$\begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix}$$

$c=s$

$d=t$

$b=-s-t$

$a=-4t-3s+2s+2t$

$=-s-2t$

$$s\begin{pmatrix} -1 \\ -1 \\ 1 \\ 0 \end{pmatrix}+t\begin{pmatrix} -2 \\ -1 \\ 0 \\ 1 \end{pmatrix}$$

col m space = span (c_1, c_2)

row space = span (R_1, R_2) = span (r_1, r_2)

null space = span $\left\{\begin{pmatrix} -1 \\ -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -2 \\ -1 \\ 0 \\ 1 \end{pmatrix}\right\}$

left null space = span $\left\{\begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}\right\}$

$$A^T \rightarrow \begin{pmatrix} 1 & 0 & 1 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \\ 4 & 1 & 5 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

P P F

$y_3 = s$

$y_2 = -s$

$y_1 = -s$

$$s\begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}$$

Dim of 4 subspaces

$A_{m \times n}$

col^m space	→ m-dim subspace	dim(col^m space) = rank
row space	→ n-dim subspace	dim(row space) = rank
null space	→ n-dim subspace	dim(null space) = n − r
left null space	→ m-dim subspace	dim(left null space) = m − r

dim(col^m space) + dim(null space) = # of col^{m}s
dim(row space) + dim(left null space) = # of col^{m}s

} Rank nullity theorem

Orthogonality of 4 subspaces

row space (A) ⊥ null space (A)
col^m space (A) ⊥ left null space (A)

row space dimension r	column space dimension r
⊥	⊥
nullspace dimension $n - r$	left nullspace $N(A^T)$ dimension $m - r$

